Modelling and control of multibody mechanical systems

Model answers

## Question 1

a) The rotation of the spherical coordinate system with respect to the fixed Cartesian coordinate system can be represented by two Euler angles $\theta$ (yaw) and $-\phi$ (pitch) in the yaw-pitch convention (note the negative sign in $\phi$ as compared to the standard convention in which the pitch rotation by the angle $\phi$ is in the opposite direction). Two single-axis-rotation transformation matrices are needed:

$$\boldsymbol{D_\theta} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is the rotation matrix by angle $\theta$ about a $z$ axis, and

$$\boldsymbol{C_{-\phi}} = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}$$

which is the rotation matrix by angle $-\phi$ about a $y$ axis. The transformation from fixed coordinates to spherical coordinates is

$$\boldsymbol{A_1} = \boldsymbol{C_{-\phi}}\boldsymbol{D_\theta} = \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix}$$

Answers

and the transformation from spherical coordinates to fixed coordinates is

$$\boldsymbol{A_2} = \boldsymbol{A_1^T} = \begin{bmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{bmatrix} \tag{1}$$

[ 8 marks ]

b) The $\boldsymbol{A_2}$ transformation in (1) can operate on the components of the $\boldsymbol{e_r}$ unit vector in the spherical coordinate system and yield the components of $\boldsymbol{e_r}$ in the fixed coordinate system, as follows:

$$\begin{bmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\phi\cos\theta \\ \cos\phi\sin\theta \\ \sin\phi \end{bmatrix}$$

and therefore in vector form

$$\boldsymbol{e_r} = \cos\theta\cos\phi\boldsymbol{i} + \sin\theta\cos\phi\boldsymbol{j} + \sin\phi\boldsymbol{k}$$

Similarly for the $\boldsymbol{e_\theta}$ unit vector

$$\begin{bmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix}$$

and therefore in vector form

$$\boldsymbol{e_\theta} = -\sin\theta\boldsymbol{i} + \cos\theta\boldsymbol{j}$$

Finally, for the $\boldsymbol{e_\phi}$ unit vector

$$\begin{bmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\phi\cos\theta \\ -\sin\phi\sin\theta \\ \cos\phi \end{bmatrix}$$

and therefore in vector form

$$\boldsymbol{e_\phi} = -\cos\theta\sin\phi\boldsymbol{i} - \sin\theta\sin\phi\boldsymbol{j} + \cos\phi\boldsymbol{k}$$

Therefore, the expressions for the spherical unit vectors, $\boldsymbol{e_r}$, $\boldsymbol{e_\theta}$, and $\boldsymbol{e_\phi}$, in terms of the fixed unit vectors are the same as those in (1.1) in Q1.

[ 4 marks ]

c) The angular velocity vector of the spherical coordinate system in spherical coordinates is

$$\boldsymbol{\Omega} = \boldsymbol{C_{-\phi}D_\theta} \begin{bmatrix} 0 \\ 0 \\ \dot\theta \end{bmatrix} + C_{-\phi} \begin{bmatrix} 0 \\ -\dot\phi \\ 0 \end{bmatrix} = \begin{bmatrix} \dot\theta\sin\phi \\ -\dot\phi \\ \dot\theta\cos\phi \end{bmatrix}$$

and therefore in vector form

$$\boldsymbol{\Omega} = \dot\theta\sin\phi\boldsymbol{e_r} - \dot\phi\boldsymbol{e_\theta} + \dot\theta\cos\phi\boldsymbol{e_\phi} \tag{2}$$

Answers

[ 4 marks ]

d) The derivatives of the spherical unit vectors are given by

$$\dot{\boldsymbol{e}}_{\boldsymbol{r}} = \boldsymbol{\Omega}\times\boldsymbol{e_r} = \left(\dot\theta\sin\phi\boldsymbol{e_r} - \dot\phi\boldsymbol{e_\theta} + \dot\theta\cos\phi\boldsymbol{e_\phi}\right)\times\boldsymbol{e_r} = \dot\theta\cos\phi\boldsymbol{e_\theta} + \dot\phi\boldsymbol{e_\phi}$$

$$\dot{\boldsymbol{e}}_{\boldsymbol{\theta}} = \boldsymbol{\Omega}\times\boldsymbol{e_\theta} = \left(\dot\theta\sin\phi\boldsymbol{e_r} - \dot\phi\boldsymbol{e_\theta} + \dot\theta\cos\phi\boldsymbol{e_\phi}\right)\times\boldsymbol{e_\theta} = -\dot\theta\cos\phi\boldsymbol{e_r} + \dot\theta\sin\phi\boldsymbol{e_\phi}$$

$$\dot{\boldsymbol{e}}_{\boldsymbol{\phi}} = \boldsymbol{\Omega}\times\boldsymbol{e_\phi} = \left(\dot\theta\sin\phi\boldsymbol{e_r} - \dot\phi\boldsymbol{e_\theta} + \dot\theta\cos\phi\boldsymbol{e_\phi}\right)\times\boldsymbol{e_\phi} = -\dot\phi\boldsymbol{e_r} - \dot\theta\sin\phi\boldsymbol{e_\theta}$$

which are the same as the expressions in (1.2) in Q1, respectively. [ 4 marks ]

## Question 2

a) The generalised coordinates of the system are the spherical coordinates (as defined in the notes) of the left wire rope, $\theta_1$ and $\phi_1$, the spherical coordinates (as defined in the notes) of the right wire rope, $\theta_2$ and $\phi_2$, and the three Euler angles of the heavy load $\psi$, $\theta$ and $\phi$; therefore, a total of 7 generalised coordinates. Note that the two radial distances associated with the spherical coordinates of the left and right wire ropes, respectively, are not used as generalised coordinates here because they are constant and equal to $l_r$. (Alternatively, the two radial distances, $r_1$ and $r_2$, can be mentioned as further two generalised coordinates but which are subject to constraints $r_1 = l_r$ and $r_2 = l_r$, respectively).

[ 2 marks ]

b) The position vector of the centre of mass of the heavy load from point $A$ is given by

$$\boldsymbol{r_C} = l_r \boldsymbol{e_{r_1}} + \frac{l_b}{2}\boldsymbol{i'} \tag{3}$$

In order to express the position vector in terms of the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$, first, the spherical unit vector $\boldsymbol{e_{r_1}}$ is written in Cartesian Earth-fixed coordinates, as in the notes,

$$\boldsymbol{e_{r_1}} = \cos\theta_1 \cos\phi_1 \boldsymbol{i} + \sin\theta_1 \cos\phi_1 \boldsymbol{j} + \sin\phi_1 \boldsymbol{k} \tag{4}$$

Then, by using the transformation from body-fixed to Earth-fixed coordinates for the yaw-pitch-roll Euler angle convention, as given in the notes,

$$\boldsymbol{A^T} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

the unit vector $\boldsymbol{i'}$ is written in Earth-fixed coordinates as follows

$$\boldsymbol{A^T} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi \\ \cos\theta\sin\psi \\ -\sin\theta \end{bmatrix}$$

and therefore in vector form

$$\boldsymbol{i'} = \cos\theta\cos\psi \boldsymbol{i} + \cos\theta\sin\psi \boldsymbol{j} - \sin\theta \boldsymbol{k} \tag{5}$$

By substituting (4) and (5) into (3), the position vector from point $A$ to point $C$ is given by

$$\boldsymbol{r_{AC}} = \left(l_r\cos\theta_1\cos\phi_1 + \frac{l_b}{2}\cos\theta\cos\psi\right)\boldsymbol{i} + \left(l_r\sin\theta_1\cos\phi_1 + \frac{l_b}{2}\cos\theta\sin\psi\right)\boldsymbol{j} + \left(l_r\sin\phi_1 - \frac{l_b}{2}\sin\theta\right)\boldsymbol{k} \tag{6}$$

[ 4 marks ]

c) By differentiation of (6), the velocity vector of the centre of mass of the heavy load is

$$\begin{aligned}\dot{\boldsymbol{r}}_{\boldsymbol{AC}} = &\left(l_r\left(-\sin\theta_1\cos\phi_1\dot{\theta}_1 - \cos\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\cos\psi\dot{\theta} - \cos\theta\sin\psi\dot{\psi}\right)\right)\boldsymbol{i} \\ &+ \left(l_r\left(\cos\theta_1\cos\phi_1\dot{\theta}_1 - \sin\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\sin\psi\dot{\theta} + \cos\theta\cos\psi\dot{\psi}\right)\right)\boldsymbol{j} \\ &+ \left(l_r\cos\phi_1\dot{\phi}_1 - \frac{l_b}{2}\cos\theta\dot{\theta}\right)\boldsymbol{k}\end{aligned}$$

[ 2 marks ]

Answers

d) The kinetic energy of the translational motion of the centre of mass of the heavy load is

$$T_t = \frac{1}{2}m\left(\left(l_r\left(-\sin\theta_1\cos\phi_1\dot{\theta}_1 - \cos\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\cos\psi\dot{\theta} - \cos\theta\sin\psi\dot{\psi}\right)\right)^2\right.$$

$$+\left(l_r\left(\cos\theta_1\cos\phi_1\dot{\theta}_1 - \sin\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\sin\psi\dot{\theta} + \cos\theta\cos\psi\dot{\psi}\right)\right)^2$$

$$\left.+\left(l_r\cos\phi_1\dot{\phi}_1 - \frac{l_b}{2}\cos\theta\dot{\theta}\right)^2\right)$$

The kinetic energy of the heavy load for its rotational motion about the centre of mass is

$$T_r = \frac{1}{2}I_{xx}\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + \frac{1}{2}I_{yy}\left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)^2 + \frac{1}{2}I_{zz}\left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)^2$$

in which the body-fixed axes components of the angular velocity vector of the heavy load, as shown in the notes, have been used. The total kinetic energy of the heavy load is therefore

$$T = T_t + T_r$$

The potential energy, with zero potential energy at the level of points $A$ and $E$, is given by

$$V = -m\boldsymbol{r_C}\cdot\boldsymbol{g} = -m\boldsymbol{r_C}\cdot(g\boldsymbol{k}) = -mg\left(l_r\sin\phi_1 - \frac{l_b}{2}\sin\theta\right)$$

The Lagrangian function is therefore

$$L = T - V = \frac{1}{2}m\left(\left(l_r\left(-\sin\theta_1\cos\phi_1\dot{\theta}_1 - \cos\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\cos\psi\dot{\theta} - \cos\theta\sin\psi\dot{\psi}\right)\right)^2\right.$$

$$+\left(l_r\left(\cos\theta_1\cos\phi_1\dot{\theta}_1 - \sin\theta_1\sin\phi_1\dot{\phi}_1\right) + \frac{l_b}{2}\left(-\sin\theta\sin\psi\dot{\theta} + \cos\theta\cos\psi\dot{\psi}\right)\right)^2$$

$$\left.+\left(l_r\cos\phi_1\dot{\phi}_1 - \frac{l_b}{2}\cos\theta\dot{\theta}\right)^2\right) + \frac{1}{2}I_{xx}\left(\dot{\phi} - \dot{\psi}\sin\theta\right)^2 + \frac{1}{2}I_{yy}\left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)^2$$

$$+\frac{1}{2}I_{zz}\left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)^2 + mg\left(l_r\sin\phi_1 - \frac{l_b}{2}\sin\theta\right)$$

[ 4 marks ]

e) The heavy load is subject to the constraint that the points $A$, $B$, $D$, and $E$ form a closed kinematic loop. Thus, position vectors around the kinematic loop are related as follows:

$$\boldsymbol{r_{ABDEA}} = \boldsymbol{r_{ABD}} - \boldsymbol{r_{AED}} = \boldsymbol{r_{AB}} + \boldsymbol{r_{BD}} - \boldsymbol{r_{ED}} - \boldsymbol{r_{AE}} = 0$$

which gives

$$l_r\boldsymbol{e_{r_1}} + l_b\boldsymbol{i'} - l_r\boldsymbol{e_{r_2}} - l_b\boldsymbol{i} = 0$$

and by expressing all the unit vectors in terms of the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$, using (4) and (5) (an expression similar to (5) can be used for $\boldsymbol{e_{r_2}}$),

$$l_r\left(\cos\theta_1\cos\phi_1\boldsymbol{i} + \sin\theta_1\cos\phi_1\boldsymbol{j} + \sin\phi_1\boldsymbol{k}\right) + l_b\left(\cos\theta\cos\psi\boldsymbol{i} + \cos\theta\sin\psi\boldsymbol{j} - \sin\theta\boldsymbol{k}\right)$$

Answers

$$-l_r\left(\cos\theta_2\cos\phi_2\boldsymbol{i}+\sin\theta_2\cos\phi_2\boldsymbol{j}+\sin\phi_2\boldsymbol{k}\right)-l_b\boldsymbol{i}=0$$

which yields the three constraint equations

$$\begin{gathered}l_r\left(\cos\theta_1\cos\phi_1-\cos\theta_2\cos\phi_2\right)+l_b\cos\theta\cos\psi-l_b=0\\ l_r\left(\sin\theta_1\cos\phi_1-\sin\theta_2\cos\phi_2\right)+l_b\cos\theta\sin\psi=0\\ l_r\left(\sin\phi_1-\sin\phi_2\right)-l_b\sin\theta=0\end{gathered}\tag{7}$$

[ 4 marks ]

f) The constraint equations can be written in differential form by differentiating (7), which gives the following three equations:

$$\begin{gathered}l_r\left(-\sin\theta_1\cos\phi_1\dot{\theta}_1-\cos\theta_1\sin\phi_1\dot{\phi}_1+\sin\theta_2\cos\phi_2\dot{\theta}_2+\cos\theta_2\sin\phi_2\dot{\phi}_2\right)\\ -l_b\left(\sin\theta\cos\psi\dot{\theta}+\cos\theta\sin\psi\dot{\psi}\right)=0\\ l_r\left(\cos\theta_1\cos\phi_1\dot{\theta}_1-\sin\theta_1\sin\phi_1\dot{\phi}_1-\cos\theta_2\cos\phi_2\dot{\theta}_2+\sin\theta_2\sin\phi_2\dot{\phi}_2\right)\\ -l_b\left(\sin\theta\sin\psi\dot{\theta}-\cos\theta\cos\psi\dot{\psi}\right)=0\\ l_r\left(\cos\phi_1\dot{\phi}_1-\cos\phi_2\dot{\phi}_2\right)-l_b\cos\theta\dot{\theta}=0\end{gathered}$$

Since there are three equations of constraint, three Lagrange multipliers, $\lambda_1$, $\lambda_2$, and $\lambda_3$, are needed. The Lagrangian equations for each of the seven generalised coordinates are

$$\begin{gathered}\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}_1}\right)-\frac{\partial L}{\partial\theta_1}-\lambda_1 l_r\sin\theta_1\cos\phi_1+\lambda_2 l_r\cos\theta_1\cos\phi_1=0\\ \frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\phi}_1}\right)-\frac{\partial L}{\partial\phi_1}-\lambda_1 l_r\cos\theta_1\sin\phi_1-\lambda_2 l_r\sin\theta_1\sin\phi_1+\lambda_3 l_r\cos\phi_1=0\\ \lambda_1 l_r\sin\theta_2\cos\phi_2-\lambda_2 l_r\cos\theta_2\cos\phi_2=0\\ \lambda_1 l_r\cos\theta_2\sin\phi_2+\lambda_2 l_r\sin\theta_2\sin\phi_2-\lambda_3 l_r\cos\phi_2=0\\ \frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}}\right)-\frac{\partial L}{\partial\theta}-\lambda_1 l_b\sin\theta\cos\psi-\lambda_2 l_b\sin\theta\sin\psi-\lambda_3 l_b\cos\theta=0\\ \frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\phi}}\right)-\frac{\partial L}{\partial\phi}=0\\ \frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\psi}}\right)-\frac{\partial L}{\partial\psi}-\lambda_1 l_b\cos\theta\sin\psi+\lambda_2 l_b\cos\theta\cos\psi=0\end{gathered}$$

Answers

[ 4 marks ]

## Question 3

a) Consider an area element on the surface of the cone of radius $r$ (the radial distance from the $Z$ axis), slant height $l$ (corresponding to a height $z$ from point $O$ along the $Z$ axis) and thickness $dl$ (corresponding to a change in height $dz$). The area of this element is

$$dA = rd\theta dl$$

In order to relate this area to the height (rather than the slant height), which is one of the cylindrical coordinates, note that

$$dz = \frac{h}{L}dl$$

Therefore, the infinitesimal area of the element in cylindrical coordinates is

$$dA = \frac{rL}{h}d\theta dz$$

[ 5 marks ]

b) i) The moment of inertia about the axis of symmetry ($Z$ axis) is:

$$I_{zz} = \int (x^2 + y^2)dm = \rho \int_A (x^2 + y^2)dA = \rho \int_A r^2 dA$$

where $\rho$ is the mass per unit area. If we consider an infinitesimal area element given in cylindrical polar coordinates then $I_{zz}$ becomes

$$I_{zz} = \frac{\rho L}{h}\int_0^h \int_0^{2\pi} r^3 d\theta dz = \frac{\rho L}{h}\int_0^h \int_0^{2\pi} \left(\frac{Rz}{h}\right)^3 d\theta dz$$

in which $r$ is a variable given by $Rz/h$, which varies from the minimum radius of 0 when $z = 0$ to the maximum radius of $R$ when $z = h$. Hence, after some effort,

$$I_{zz} = \frac{\rho\pi L R^3}{2}$$

Finally, the mass of the cone is given by

$$m = \rho A = \rho\pi RL$$

and therefore $I_{zz}$ becomes

$$I_{zz} = \frac{1}{2}mR^2$$

[ 7 marks ]

ii) The moment of inertia about the $X$ axis can be found similarly via the equation

$$I_{xx} = \int (y^2 + z^2)dm = \rho \int_A (y^2 + z^2)dA = \rho \int_A (r^2\cos^2\theta + z^2)dA$$

By using a similar cylindrical polar area element as above the area integral becomes

$$I_{xx} = \frac{\rho L}{h}\int_0^h \int_0^{2\pi} (r^2\cos^2\theta + z^2) r d\theta dz = \frac{\rho L}{h}\int_0^h \int_0^{2\pi} \left(\left(\frac{Rz}{h}\right)^2 \cos^2\theta + z^2\right)\left(\frac{Rz}{h}\right) d\theta dz$$

Answers

and after some effort

$$I_{xx} = \frac{1}{4}\rho\pi RLh\left(R^2 + 2h^2\right)$$

and by making use of the mass expression of the cone

$$I_{xx} = \frac{1}{4}m\left(R^2 + 2h^2\right)$$

[ 6 marks ]

iii) Due to the symmetry of the cone $I_{yy}$ is the same as $I_{xx}$, i.e.

$$I_{yy} = I_{xx}$$

[ 2 marks ]

Answers

## Question 4

a)

$$\dot{\boldsymbol{b}} = \boldsymbol{\Omega_b} \times \boldsymbol{b}$$

$$\ddot{\boldsymbol{b}} = \dot{\boldsymbol{\Omega}}_{\boldsymbol{b}} \times \boldsymbol{b} + \boldsymbol{\Omega_b} \times \dot{\boldsymbol{b}} = \dot{\boldsymbol{\Omega}}_{\boldsymbol{b}} \times \boldsymbol{b} + \boldsymbol{\Omega_b} \times (\boldsymbol{\Omega_b} \times \boldsymbol{b})$$

[ 2 marks ]

b) The angular velocity vector of the heavy load with respect to Earth is given in the notes (roll-pitch-yaw convention) using body fixed coordinates, as follows:

$$\boldsymbol{\Omega} = \begin{bmatrix} \dot{\phi} - \dot{\psi}\sin\theta \\ \dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi \\ -\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi \end{bmatrix} \tag{8}$$

and therefore in vector form it is

$$\boldsymbol{\Omega} = \left(\dot{\phi} - \dot{\psi}\sin\theta\right)\boldsymbol{i'} + \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)\boldsymbol{j'} + \left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)\boldsymbol{k'} \tag{9}$$

The heavy load angular acceleration vector with respect to Earth in the same (heavy-load-fixed) coordinate system is therefore obtained by differentiating (9) as follows:

$$\dot{\boldsymbol{\Omega}} = \frac{d\boldsymbol{\Omega}}{dt} = \frac{d'\boldsymbol{\Omega}}{dt} + \boldsymbol{\Omega} \times \boldsymbol{\Omega} = \frac{d'\boldsymbol{\Omega}}{dt} =$$

$$\left(\ddot{\phi} - \ddot{\psi}\sin\theta - \dot{\psi}\dot{\theta}\cos\theta\right)\boldsymbol{i'} + \left(\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi + \ddot{\psi}\cos\theta\sin\phi - \dot{\psi}\dot{\theta}\sin\theta\sin\phi + \dot{\psi}\dot{\phi}\cos\theta\cos\phi\right)\boldsymbol{j'}$$

$$+ \left(-\ddot{\theta}\sin\phi - \dot{\theta}\dot{\phi}\cos\phi + \ddot{\psi}\cos\theta\cos\phi - \dot{\psi}\dot{\theta}\sin\theta\cos\phi - \dot{\psi}\dot{\phi}\cos\theta\sin\phi\right)\boldsymbol{k'}$$

Answers

[ 3 marks ]

c) The position vector from point $B$ to point $C$ (heavy load centre of mass) is

$$\boldsymbol{r_{BC}} = \frac{l_b}{2}\boldsymbol{i'} \tag{10}$$

The following calculations are required

$$\dot{\boldsymbol{i}}' = \boldsymbol{\Omega} \times \boldsymbol{i'} = \left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)\boldsymbol{j'} - \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)\boldsymbol{k'}$$

$$\dot{\boldsymbol{j}}' = \boldsymbol{\Omega} \times \boldsymbol{j'} = -\left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)\boldsymbol{i'} + \left(\dot{\phi} - \dot{\psi}\sin\theta\right)\boldsymbol{k'}$$

$$\dot{\boldsymbol{k}}' = \boldsymbol{\Omega} \times \boldsymbol{k'} = \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)\boldsymbol{i'} - \left(\dot{\phi} - \dot{\psi}\sin\theta\right)\boldsymbol{j'}$$

The relative velocity vector is obtained by differentiating (10), as follows:

$$\dot{\boldsymbol{r}}_{\boldsymbol{BC}} = \frac{l_b}{2}\boldsymbol{\Omega} \times \boldsymbol{i'} = \frac{l_b}{2}\left(\left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)\boldsymbol{j'} - \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)\boldsymbol{k'}\right)$$

By a further differentiation, the relative acceleration vector is obtained as

$$\ddot{\boldsymbol{r}}_{\boldsymbol{BC}} = \frac{l_b}{2}\left(\left(-\ddot{\theta}\sin\phi - \dot{\theta}\dot{\phi}\cos\phi + \ddot{\psi}\cos\theta\cos\phi - \dot{\psi}\dot{\theta}\sin\theta\cos\phi - \dot{\psi}\dot{\phi}\cos\theta\sin\phi\right)\boldsymbol{j'}\right.$$

$$+\left(-\dot{\theta}\sin\phi+\dot{\psi}\cos\theta\cos\phi\right)\boldsymbol{\Omega}\times\boldsymbol{j'}$$
$$-\left(\ddot{\theta}\cos\phi-\dot{\theta}\dot{\phi}\sin\phi+\ddot{\psi}\cos\theta\sin\phi-\dot{\psi}\dot{\theta}\sin\theta\sin\phi+\dot{\psi}\dot{\phi}\cos\theta\cos\phi\right)\boldsymbol{k'}$$
$$-\left(\dot{\theta}\cos\phi+\dot{\psi}\cos\theta\sin\phi\right)\boldsymbol{\Omega}\times\boldsymbol{k'}\Big)$$

$$\begin{aligned}=\frac{l_b}{2}\Big(\Big(&-\ddot{\theta}\sin\phi-\dot{\theta}\dot{\phi}\cos\phi+\ddot{\psi}\cos\theta\cos\phi-\dot{\psi}\dot{\theta}\sin\theta\cos\phi-\dot{\psi}\dot{\phi}\cos\theta\sin\phi\Big)\boldsymbol{j'}\\+\Big(&-\dot{\theta}\sin\phi+\dot{\psi}\cos\theta\cos\phi\Big)\Big(-\Big(-\dot{\theta}\sin\phi+\dot{\psi}\cos\theta\cos\phi\Big)\boldsymbol{i'}+\Big(\dot{\phi}-\dot{\psi}\sin\theta\Big)\boldsymbol{k'}\Big)\\&-\Big(\ddot{\theta}\cos\phi-\dot{\theta}\dot{\phi}\sin\phi+\ddot{\psi}\cos\theta\sin\phi-\dot{\psi}\dot{\theta}\sin\theta\sin\phi+\dot{\psi}\dot{\phi}\cos\theta\cos\phi\Big)\boldsymbol{k'}\\&-\Big(\dot{\theta}\cos\phi+\dot{\psi}\cos\theta\sin\phi\Big)\Big(\Big(\dot{\theta}\cos\phi+\dot{\psi}\cos\theta\sin\phi\Big)\boldsymbol{i'}-\Big(\dot{\phi}-\dot{\psi}\sin\theta\Big)\boldsymbol{j'}\Big)\Big)\end{aligned}\qquad(11)$$

The relative accceleration vector can also be obtained by using the results from parts a) and b) and the vector triple product identity, as follows:

$$\ddot{\boldsymbol{r}}_{\boldsymbol{BC}}=\frac{l_b}{2}\left(\dot{\boldsymbol{\Omega}}\times\boldsymbol{i'}+\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\boldsymbol{i'})\right)=\frac{l_b}{2}\left(\dot{\boldsymbol{\Omega}}\times\boldsymbol{i'}+(\boldsymbol{\Omega}\cdot\boldsymbol{i'})\,\boldsymbol{\Omega}-(\boldsymbol{\Omega}\cdot\boldsymbol{\Omega})\,\boldsymbol{i'}\right)$$

[ 3 marks ]

d) The acceleration vector of the heavy load centre of mass with respect to Earth, $\ddot{\boldsymbol{r}}_{\boldsymbol{AC}}$, is given by

$$\ddot{\boldsymbol{r}}_{\boldsymbol{AC}}=\ddot{\boldsymbol{r}}_{\boldsymbol{AB}}+\ddot{\boldsymbol{r}}_{\boldsymbol{BC}}$$

where $\ddot{\boldsymbol{r}}_{\boldsymbol{AB}}$ is the acceleration vector of point $B$ on the left wire rope with respect to Earth, which in spherical coordinates, as given in the notes with $r=l_r$, is

$$\ddot{\boldsymbol{r}}_{\boldsymbol{AB}}=(-l_r\dot{\theta}_1^2\cos^2\phi_1-l_r\dot{\phi}_1^2)\boldsymbol{e}_{\boldsymbol{r_1}}+(l_r\ddot{\theta}_1\cos\phi_1-2l_r\dot{\theta}_1\dot{\phi}_1\sin\phi_1)\boldsymbol{e}_{\boldsymbol{\theta_1}}+(l_r\dot{\theta}_1^2\sin\phi_1\cos\phi_1+l_r\ddot{\phi}_1)\boldsymbol{e}_{\boldsymbol{\phi_1}}$$

Therefore,by also using (11)

$$\ddot{\boldsymbol{r}}_{\boldsymbol{AC}}=(-l_r\dot{\theta}_1^2\cos^2\phi_1-l_r\dot{\phi}_1^2)\boldsymbol{e}_{\boldsymbol{r_1}}+(l_r\ddot{\theta}_1\cos\phi_1-2l_r\dot{\theta}_1\dot{\phi}_1\sin\phi_1)\boldsymbol{e}_{\boldsymbol{\theta_1}}+(l_r\dot{\theta}_1^2\sin\phi_1\cos\phi_1+l_r\ddot{\phi}_1)\boldsymbol{e}_{\boldsymbol{\phi_1}}$$

$$\begin{aligned}+\frac{l_b}{2}\Big(\Big(&-\ddot{\theta}\sin\phi-\dot{\theta}\dot{\phi}\cos\phi+\ddot{\psi}\cos\theta\cos\phi-\dot{\psi}\dot{\theta}\sin\theta\cos\phi-\dot{\psi}\dot{\phi}\cos\theta\sin\phi\Big)\boldsymbol{j'}\\+\Big(&-\dot{\theta}\sin\phi+\dot{\psi}\cos\theta\cos\phi\Big)\Big(-\Big(-\dot{\theta}\sin\phi+\dot{\psi}\cos\theta\cos\phi\Big)\boldsymbol{i'}+\Big(\dot{\phi}-\dot{\psi}\sin\theta\Big)\boldsymbol{k'}\Big)\\&-\Big(\ddot{\theta}\cos\phi-\dot{\theta}\dot{\phi}\sin\phi+\ddot{\psi}\cos\theta\sin\phi-\dot{\psi}\dot{\theta}\sin\theta\sin\phi+\dot{\psi}\dot{\phi}\cos\theta\cos\phi\Big)\boldsymbol{k'}\\&-\Big(\dot{\theta}\cos\phi+\dot{\psi}\cos\theta\sin\phi\Big)\Big(\Big(\dot{\theta}\cos\phi+\dot{\psi}\cos\theta\sin\phi\Big)\boldsymbol{i'}-\Big(\dot{\phi}-\dot{\psi}\sin\theta\Big)\boldsymbol{j'}\Big)\Big)\end{aligned}$$

The unit vectors $\boldsymbol{e}_{\boldsymbol{r_1}}$, $\boldsymbol{e}_{\boldsymbol{\theta_1}}$, and $\boldsymbol{e}_{\boldsymbol{\phi_1}}$ can be transformed to Earth-fixed coordinates as given in the notes, as follows:

$$\begin{aligned}\boldsymbol{e}_{\boldsymbol{r_1}}&=\cos\theta_1\cos\phi_1\boldsymbol{i}+\sin\theta_1\cos\phi_1\boldsymbol{j}+\sin\phi_1\boldsymbol{k}\\\boldsymbol{e}_{\boldsymbol{\theta_1}}&=-\sin\theta_1\boldsymbol{i}+\cos\theta_1\boldsymbol{j}\\\boldsymbol{e}_{\boldsymbol{\phi_1}}&=-\cos\theta_!\sin\phi_1\boldsymbol{i}-\sin\theta_1\sin\phi_1\boldsymbol{j}+\cos\phi_1\boldsymbol{k}\end{aligned}\qquad(12)$$

and the unit vectors $\boldsymbol{i'}$, $\boldsymbol{j'}$, and $\boldsymbol{k'}$ can be transformed to Earth-fixed coordinates using the transformation from body-fixed to Earth-fixed coordinates for the yaw-pitch-roll Euler angle convention, as given in the notes.

[ 3 marks ]

Answers

e) Earth, the two wire ropes and the heavy load form a closed kinematic loop. Therefore, the position vectors around that loop can be related as follows:

$$l_r \boldsymbol{e_{r_1}} + l_b \boldsymbol{i'} - l_r \boldsymbol{e_{r_2}} - l_b \boldsymbol{i} = 0$$

and by expressing all the unit vectors in terms of the Earth-fixed unit vectors $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$, using (12) for $\boldsymbol{e_{r_1}}$ (an expression similar to (12) can be used for $\boldsymbol{e_{r_2}}$) and body-fixed to Earth-fixed transformations for the yaw-pitch-roll Euler angle convention given in the notes for $\boldsymbol{i'}$,

$$l_r \left(\cos\theta_1 \cos\phi_1 \boldsymbol{i} + \sin\theta_1 \cos\phi_1 \boldsymbol{j} + \sin\phi_1 \boldsymbol{k}\right) + l_b \left(\cos\theta \cos\psi \boldsymbol{i} + \cos\theta \sin\psi \boldsymbol{j} - \sin\theta \boldsymbol{k}\right)$$

$$-l_r \left(\cos\theta_2 \cos\phi_2 \boldsymbol{i} + \sin\theta_2 \cos\phi_2 \boldsymbol{j} + \sin\phi_2 \boldsymbol{k}\right) - l_b \boldsymbol{i} = 0$$

which yields the three constraint equations

$$\begin{gathered} l_r \left(\cos\theta_1 \cos\phi_1 - \cos\theta_2 \cos\phi_2\right) + l_b \cos\theta \cos\psi - l_b = 0 \\ l_r \left(\sin\theta_1 \cos\phi_1 - \sin\theta_2 \cos\phi_2\right) + l_b \cos\theta \sin\psi = 0 \\ l_r \left(\sin\phi_1 - \sin\phi_2\right) - l_b \sin\theta = 0 \end{gathered} \tag{13}$$

[ 3 marks ]

f) Two unknown constraint forces, one along each of the wire ropes, are acting on the heavy load. Newton's second law of motion applied on the heavy load therefore gives

$$F_1 \boldsymbol{e_{r_1}} + F_2 \boldsymbol{e_{r_2}} + mg\boldsymbol{k} = m\left(a_x \boldsymbol{i} + a_y \boldsymbol{j} + a_z \boldsymbol{k}\right)$$

where $F_1$ and $F_2$ are the magnitudes of the unknown left and right wire rope constraint forces, respectively. By using (12) to write $\boldsymbol{e_{r_1}}$ in Earth-fixed coordinates (and a similar expression for $\boldsymbol{e_{r_2}}$), the above equation can be written as

$$F_1 \left(\cos\theta_1 \cos\phi_1 \boldsymbol{i} + \sin\theta_1 \cos\phi_1 \boldsymbol{j} + \sin\phi_1 \boldsymbol{k}\right) + F_2 \left(\cos\theta_2 \cos\phi_2 \boldsymbol{i} + \sin\theta_2 \cos\phi_2 \boldsymbol{j} + \sin\phi_2 \boldsymbol{k}\right)$$

$$+mg\boldsymbol{k} = m\left(a_x \boldsymbol{i} + a_y \boldsymbol{j} + a_z \boldsymbol{k}\right)$$

Therefore, three scalar equations can be written, as follows:

$$\begin{gathered} F_1 \cos\theta_1 \cos\phi_1 + F_2 \cos\theta_2 \cos\phi_2 - ma_x = 0 \\ F_1 \sin\theta_1 \cos\phi_1 + F_2 \sin\theta_2 \cos\phi_2 - ma_y = 0 \\ F_1 \sin\phi_1 + F_2 \sin\phi_2 + mg - ma_z = 0 \end{gathered} \tag{14}$$

[ 3 marks ]

g) The rotational motion of the heavy load about its centre of mass $C$, according to $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$ (the Euler equations in the notes can be used directly), should also be considered. The moments acting on the heavy load with respect the centre of mass $C$ are

$$\boldsymbol{N} = -\frac{l_b}{2} \boldsymbol{i'} \times F_1 \boldsymbol{e_{r_1}} + \frac{l_b}{2} \boldsymbol{i'} \times F_2 \boldsymbol{e_{r_2}}$$

By writing all the unit vectors in Earth-fixed coordinates, using (12) for $\boldsymbol{e_{r_1}}$ (and a similar expression for $\boldsymbol{e_{r_2}}$) and body-fixed to Earth-fixed transformations for the yaw-pitch-roll Euler angle convention given in the notes for $\boldsymbol{i'}$, the cross products can be

Answers

calculated. Then by using Earth-fixed to body-fixed transformations for the yaw-pitch-roll Euler angle convention given in the notes, the moment vector can be written in heavy-load-fixed coordinates as follows:

$$\boldsymbol{N} = N_1\boldsymbol{i'} + N_2\boldsymbol{j'} + N_3\boldsymbol{k'}$$

where $N_1$, $N_2$, and $N_3$ are functions of the generalised coordinates and the unknown constraint forces $F_1$ and $F_2$. The Euler equations of motion as given in the notes are therefore

$$\begin{aligned} I_{xx}\dot{\Omega}_x - \Omega_y\Omega_z(I_{yy} - I_{zz}) &= N_1 \\ I_{yy}\dot{\Omega}_y - \Omega_z\Omega_x(I_{zz} - I_{xx}) &= N_2 \\ I_{zz}\dot{\Omega}_z - \Omega_x\Omega_y(I_{xx} - I_{yy}) &= N_3 \end{aligned}$$

with the components of the angular velocity vector as given in (8) substituted

$$\begin{aligned} I_{xx}\left(\ddot{\phi} - \ddot{\psi}\sin\theta - \dot{\psi}\dot{\theta}\cos\theta\right) - \left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)\left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)(I_{yy} - I_{zz}) = N_1 \\ I_{yy}\left(\ddot{\theta}\cos\phi - \dot{\theta}\dot{\phi}\sin\phi + \ddot{\psi}\cos\theta\sin\phi - \dot{\psi}\dot{\theta}\sin\theta\sin\phi + \dot{\psi}\dot{\phi}\cos\theta\cos\phi\right) \\ - \left(-\dot{\theta}\sin\phi + \dot{\psi}\cos\theta\cos\phi\right)\left(\dot{\phi} - \dot{\psi}\sin\theta\right)(I_{zz} - I_{xx}) = N_2 \\ I_{zz}\left(-\ddot{\theta}\sin\phi - \dot{\theta}\dot{\phi}\cos\phi + \ddot{\psi}\cos\theta\cos\phi - \dot{\psi}\dot{\theta}\sin\theta\cos\phi - \dot{\psi}\dot{\phi}\cos\theta\sin\phi\right) \\ - \left(\dot{\phi} - \dot{\psi}\sin\theta\right)\left(\dot{\theta}\cos\phi + \dot{\psi}\cos\theta\sin\phi\right)(I_{xx} - I_{yy}) = N_3 \end{aligned} \tag{15}$$

Answers

The system has 7 generalised coordinates ($\theta_1$, $\phi_1$, $\theta_2$, $\phi_2$, $\psi$, $\theta$ and $\phi$) and 3 constraints, as given in (13), therefore it has 4 degrees of freedom. In order to solve its motion, the 3 equations found by considering the translational motion of the heavy load centre of mass in (14), the 3 Euler equations of motion found by considering the rotational motion about the centre of mass of the heavy load in (15), and the 3 constraint equations in (13), that is a total of 9 equations with 9 unknown variables (7 generalised coordinates and 2 unknown forces $F_1$ and $F_2$), can be solved simultaneously.

[ 3 marks ]